Ballistic thermal transport

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Contents

Materials

Two temperature model

Model Equations

Steady state analysis

Effective thermal conductivity

Dynamic Approach

Results

Material Suggestions



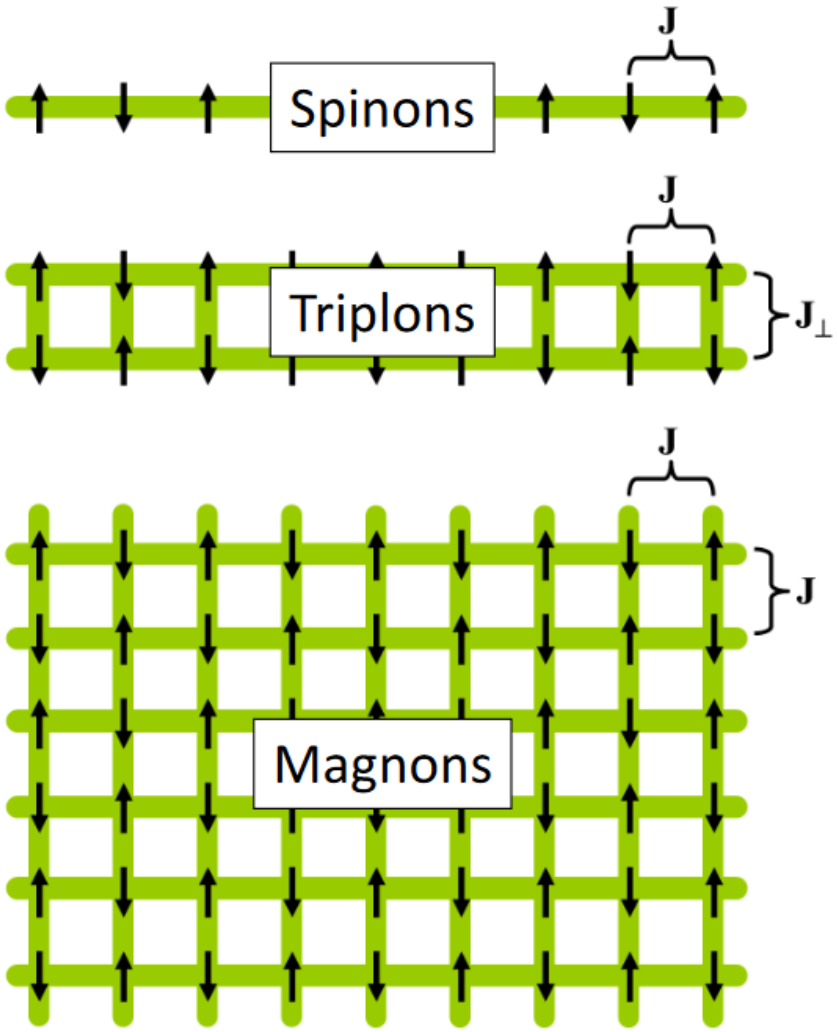
Materials

Antiferromagnetic Heisenberg spin chain , ladder materials.

- Spin chain (SC) : Ballistic thermal transport
- Spin ladder (SL) : Diffusive thermal transport



Carriers





Two temperature model

$$C_p \partial_t T_p = k_p \partial_x^2 T_p - g(T_p - T_m)$$
$$C_m \partial_t T_m = k_m \partial_x^2 T_m - g(T_m - T_p) \quad (1)$$

Fails to model ballistic thermal transport.



Equilibrium

$$\frac{\partial \Delta T}{\partial t} = -\frac{\Delta T}{\tau}, \quad \Delta T = T_p - T_m, T_m = \frac{T_+ + T_-}{2} \tag{2}$$

The individual contributions of the two subsystems are

$$\begin{aligned} \frac{\partial T_p}{\partial t} &= \frac{c_m}{C}\frac{T_m - T_p}{\tau} \\ \frac{\partial T_+}{\partial t} &= \frac{c_p}{C}\frac{T_p - T_+}{\tau} \\ \frac{\partial T_-}{\partial t} &= \frac{c_p}{C}\frac{T_p - T_-}{\tau} \end{aligned} \tag{3}$$



Spatial dependence

$$\begin{aligned} \frac{\partial \epsilon_p}{\partial t} &= D\frac{\partial^2 \epsilon_p}{\partial x^2} + \frac{c_p c_m}{C}\frac{T_m - T_p}{\tau} \\ \frac{\partial \epsilon_\pm}{\partial t} \pm v\frac{\partial \epsilon_\pm}{\partial x} &= \frac{c_p c_\pm}{C}\frac{T_p - T_\pm}{\tau} \end{aligned} \tag{4}$$

- Phonon energy current $Q_p = -\kappa_p \frac{\partial T_p}{\partial x}$
- Magnetic energy current $Q_m = v\epsilon_+ - v\epsilon_-$



Temperature equations

$$\frac{\partial T_p}{\partial t} = D\frac{\partial^2 T_p}{\partial x^2} + \frac{c_m}{C}\frac{T_m - T_p}{\tau}$$
$$\frac{\partial T_\pm}{\partial t} \pm v\frac{\partial T_\pm}{\partial x} = \frac{c_p}{C}\frac{T_p - T_\pm}{\tau} \tag{5}$$

$$Q = Q_p + Q_m = -\kappa_p\frac{\partial T_p}{\partial x} + v\frac{c_m}{2}(T_+ - T_-). \tag{6}$$



Steady state equations

$$v\frac{\partial T_m}{\partial x} = -\frac{c_p}{C}\frac{\Delta T_m}{\tau}$$
$$D\frac{\partial^2 T_p}{\partial x^2} + \frac{c_m}{C}\frac{T_m - T_p}{\tau} = 0 \qquad (7)$$

$$Q = -\kappa_p\frac{\partial T_p}{\partial x} + vc_m\Delta T_m \quad , \quad \Delta T_m = \frac{T_+ - T_-}{2}. \qquad (8)$$

- System $-L/2 < x < L/2$
- $Q|_{x=\pm L/2} = Q_p$
- $Q_m|_{x=\pm L/2} = (T_+ - T_-)|_{x=\pm L/2} = 0$

Solving (8) for ΔT_m and substituting in (7) we get



$$\frac{\partial T_m}{\partial x} = \frac{c_p}{c_m}\frac{1}{Cv^2\tau}(Q + \kappa_p \frac{\partial T_p}{\partial x})$$

$\kappa = Cv^2\tau$ defines an effective "transfer thermal conductivity"

$$\frac{\partial T_m}{\partial x} = -\frac{c_p}{c_m}\frac{1}{\kappa}(Q + \kappa_p \frac{\partial T_p}{\partial x}) \quad (9)$$



- c_p , c_m are temperature independent
- $T_p(x = 0) = T_m(0) = T_0$

$$T_m = T_0 - \frac{c_p}{c_m}\frac{1}{\kappa}Qx - \frac{c_p}{c_m}\frac{\kappa_p}{\kappa}(T_p - T_0) \tag{10}$$



- With $\partial(T_p - T_0)/\partial x = -Q/\kappa_p$

$$T_p = T_0 - (\frac{Bx}{A^2} - \frac{(B/A^2 - 1/\kappa_p)\sinh Ax}{A\cosh AL/2})Q \qquad (11)$$

$$A^2 = \frac{1}{C\tau}(\frac{c_p c_m}{\kappa_p} + \frac{c_p^2}{\kappa})$$

$$B = \frac{1}{C\tau}\frac{c_p^2}{\kappa\kappa_p}.$$



Limits

If $\tau \to 0$ then $A \to \infty$.At this limit the last term of eq. (11) tends to zero , so

$$T_p(x) \to T_0 - \frac{BQ}{A^2}x$$

For a two-diffusion model

$$T_p(x) \to T_0 - \frac{Q}{(\kappa_m + \kappa_p)}x$$



- SC : $SrCuO_2$
- SL : $Ca_9La_5Cu_24O_2$

Quantity	Spin ladder	Spin chain	Dimensions
c_m	$2,8 \times 10^6$	$2,8 \times 10^6$	$JK^{-1}m^{-3}$
c_p	$1,5 \times 10^5$	3×10^4	$JK^{-1}m^{-3}$
κ_p	1	8	$WK^{-1}m^{-1}$
κ_m	45	40	$WK^{-1}m^{-1}$
τ	4×10^{-4}	10^{-12}	s
v	–	2×10^4	ms^{-1}



Comparison

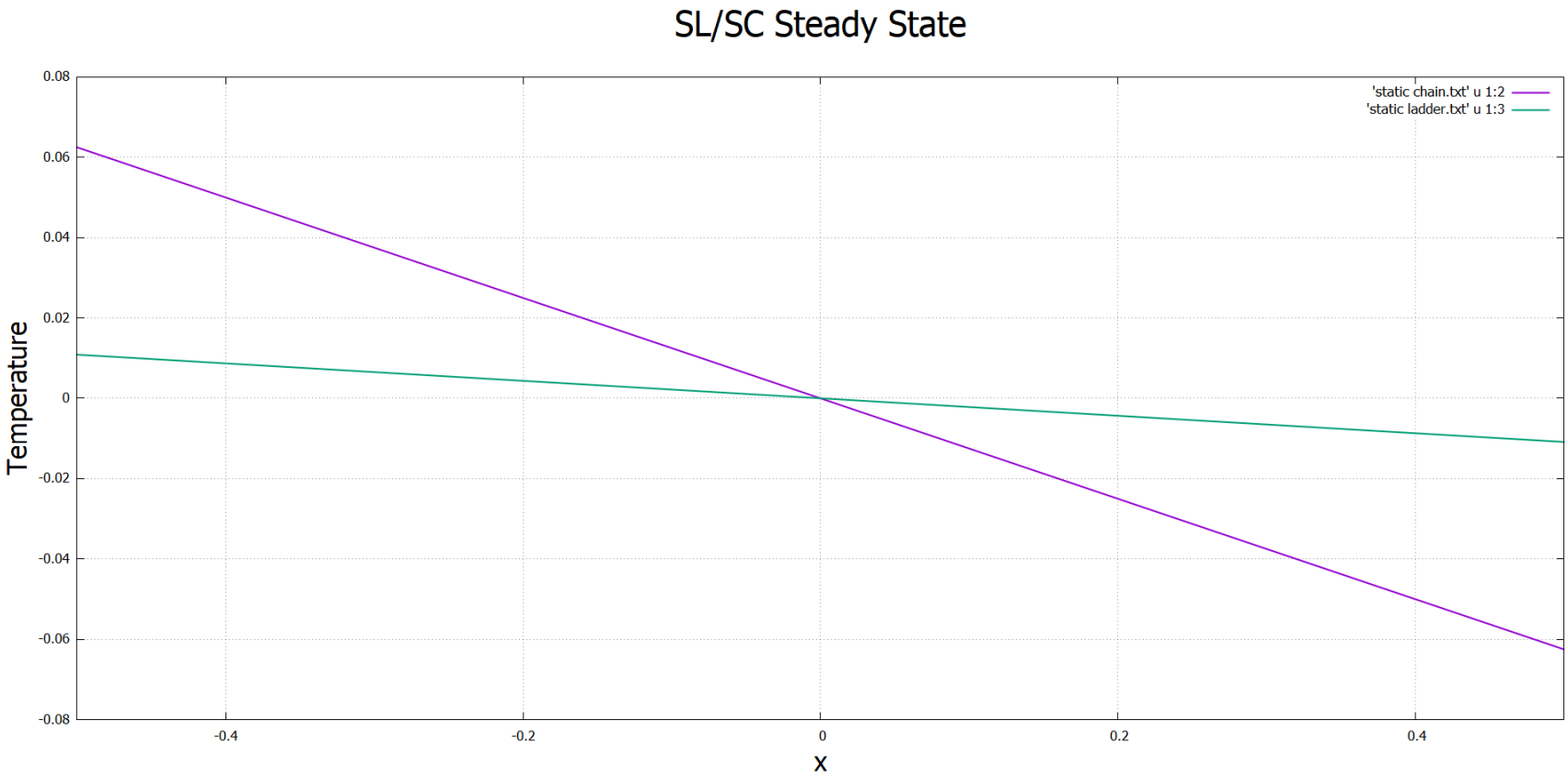


Figure: Green : Spin ladder / Purple : Spin Chain



Effective thermal conductivity

$$\kappa_{eff} = -QL/\Delta T_p = \frac{1}{\frac{B}{A^2} - \frac{B/A^2 - 1/\kappa_p}{AL}\tanh(AL/2)} \tag{12}$$

$$\Delta T_p = T(L/2) - T(-L/2)$$



Approximation

- $c_m \sim T_0/v$
- $c_p \sim T_0/v_p$
- $v_p << v$.

$$\begin{aligned} \frac{B}{A^2} &= \frac{1}{\kappa_p + \kappa\frac{c_m}{c_p}} \sim \frac{1}{T_0(l_p + l)} \\ \frac{1}{\kappa_p} &\sim \frac{1}{T_0 l_p}, \end{aligned}$$

with $c_m << c_p$, $\kappa_p = c_p v_p l_p$ and $l = v\tau$.

$$\begin{aligned} AL &\sim \sqrt{\frac{L^2}{l}(\frac{1}{l} + \frac{1}{l_p})} \\ \kappa_{eff} &\sim T_0(l + l_p) \sim \kappa_p + \kappa, \quad AL \to \infty \end{aligned} \tag{13}$$



Dynamic temperature equations

$$\frac{\partial T_p}{\partial t} = D\frac{\partial^2 T_p}{\partial x^2} + \frac{c_m}{C}\frac{T_m - T_p}{\tau}$$

$$\frac{\partial T_\pm}{\partial t} \pm v\frac{\partial T_\pm}{\partial x} = \frac{c_p}{C}\frac{T_p - T_\pm}{\tau}$$

- System of length L
- $Q_p = 0 \Rightarrow \frac{\partial T_p}{\partial x}|_{x=0,L} = 0$
- $Q_m = 0 \Rightarrow T_+|_{x=0,L} = T_-|_{x=0,L}$



Fourier approximation

$$T_p = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos q_n x, \quad q_n = \frac{\pi n}{L}$$

$$T_\pm = \frac{b_0}{2} + \sum_{n=1}^{+\infty} b_n \cos q_n x \pm c_n \sin q_n x. \tag{14}$$

and substituting (14) above we get

$$a_0(t) = a_0(0) - \frac{c_m}{C}(a_0 - b_0)|_0(1 - e^{-t/\tau})$$

$$b_0(t) = b_0(0) - \frac{c_p}{C}(b_0 - a_0)|_0(1 - e^{-t/\tau}). \tag{15}$$

for finite wavevector q_n,

$$\dot{a}_n + Dq_n^2 a_n + \frac{c_m}{C\tau}(a_n - b_n) = 0$$

$$\dot{b}_n + vq_n c_n + \frac{c_p}{C\tau}(b_n - a_n) = 0$$

$$\dot{c}_n - vq_n b_n + \frac{c_p}{C\tau} c_n = 0, \tag{16}$$



Varying v, $D = 0.001$, $c_p = 0.9$, $c_m = 0.1$

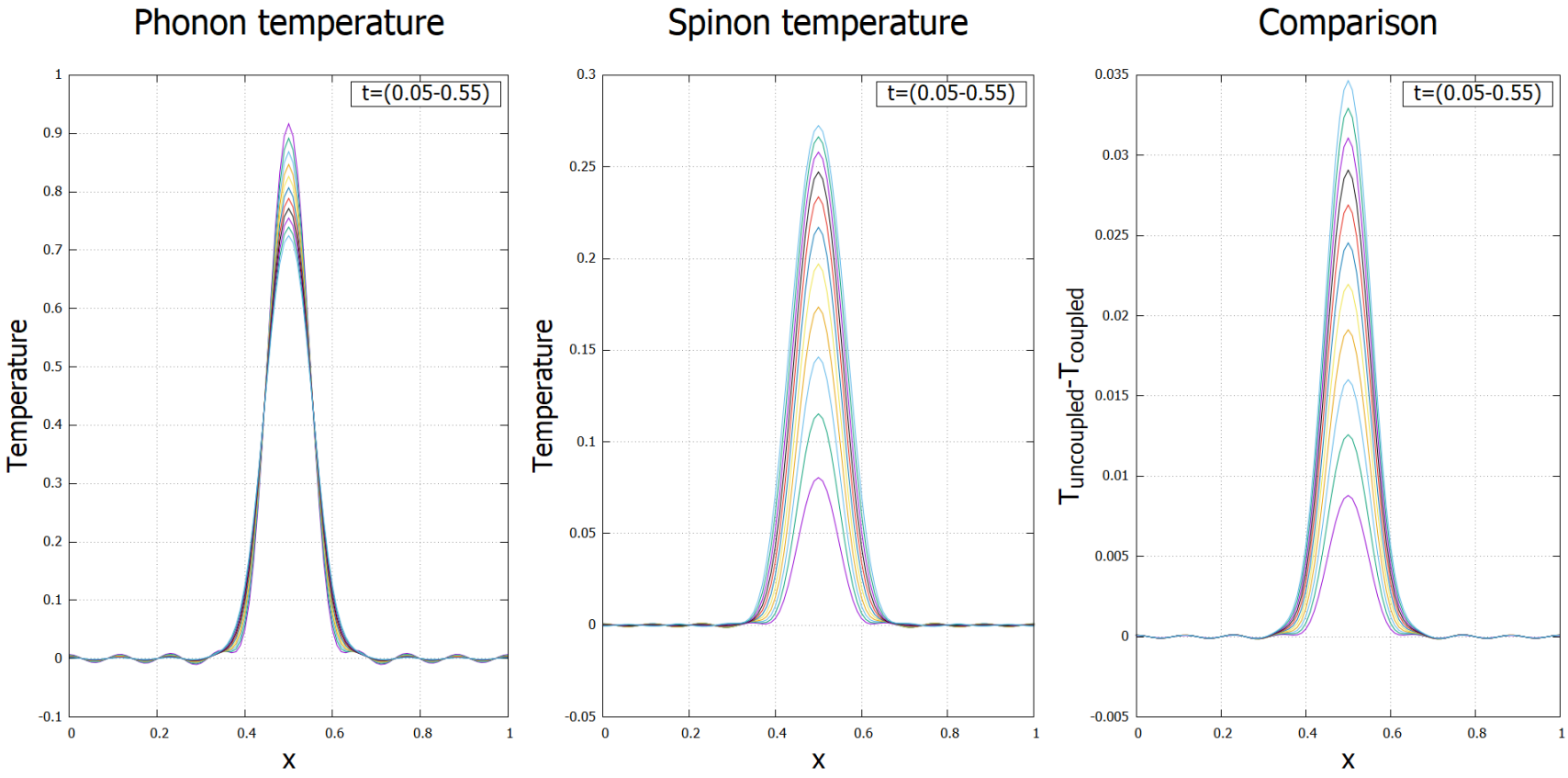


Figure: $v = 0.1$

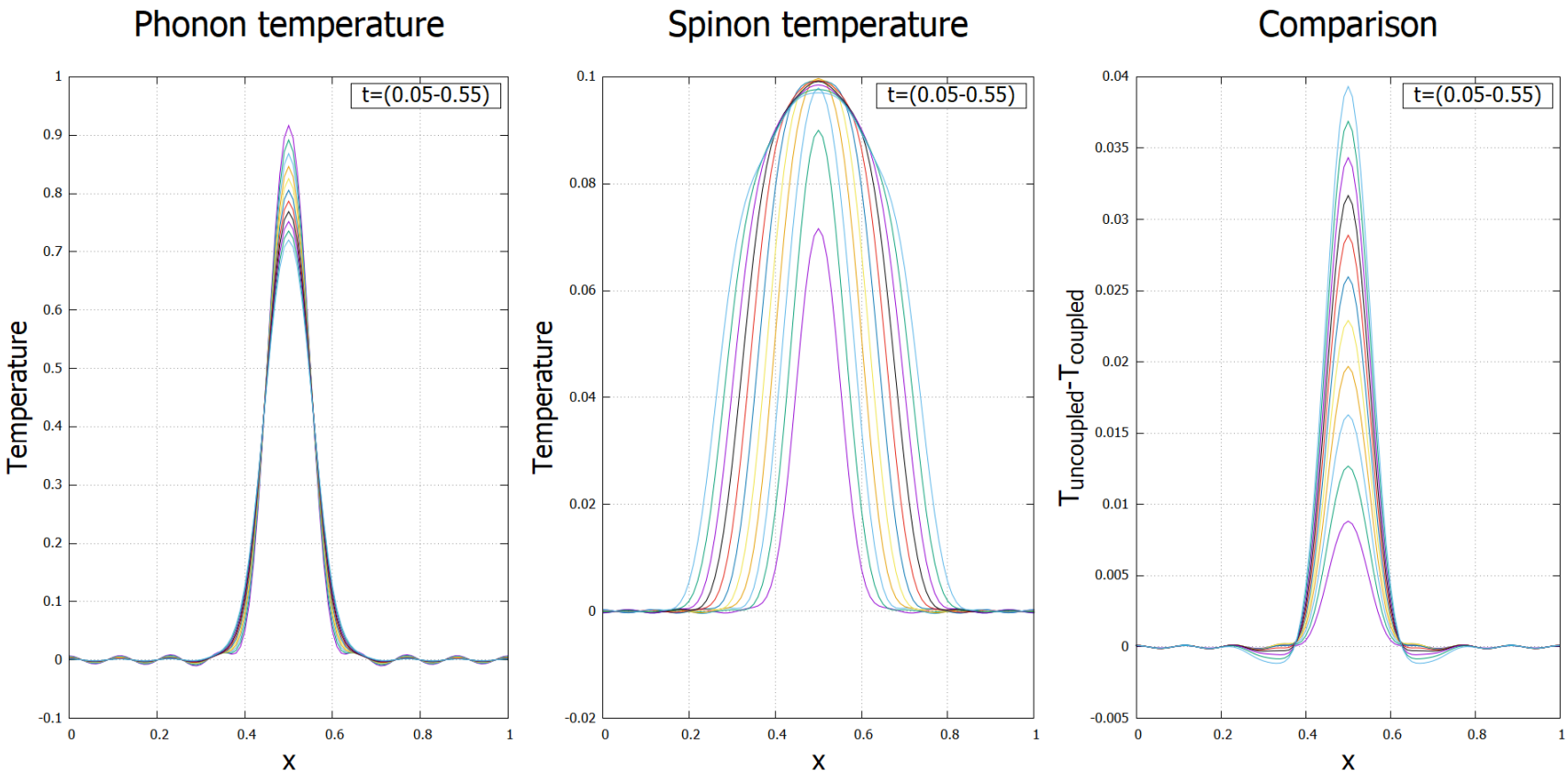

Phonon temperature
Spinon temperature
Comparison
t=(0.05-0.55)
Temperature
T_uncoupled-T_coupled
x

Figure: $v = 0.4$

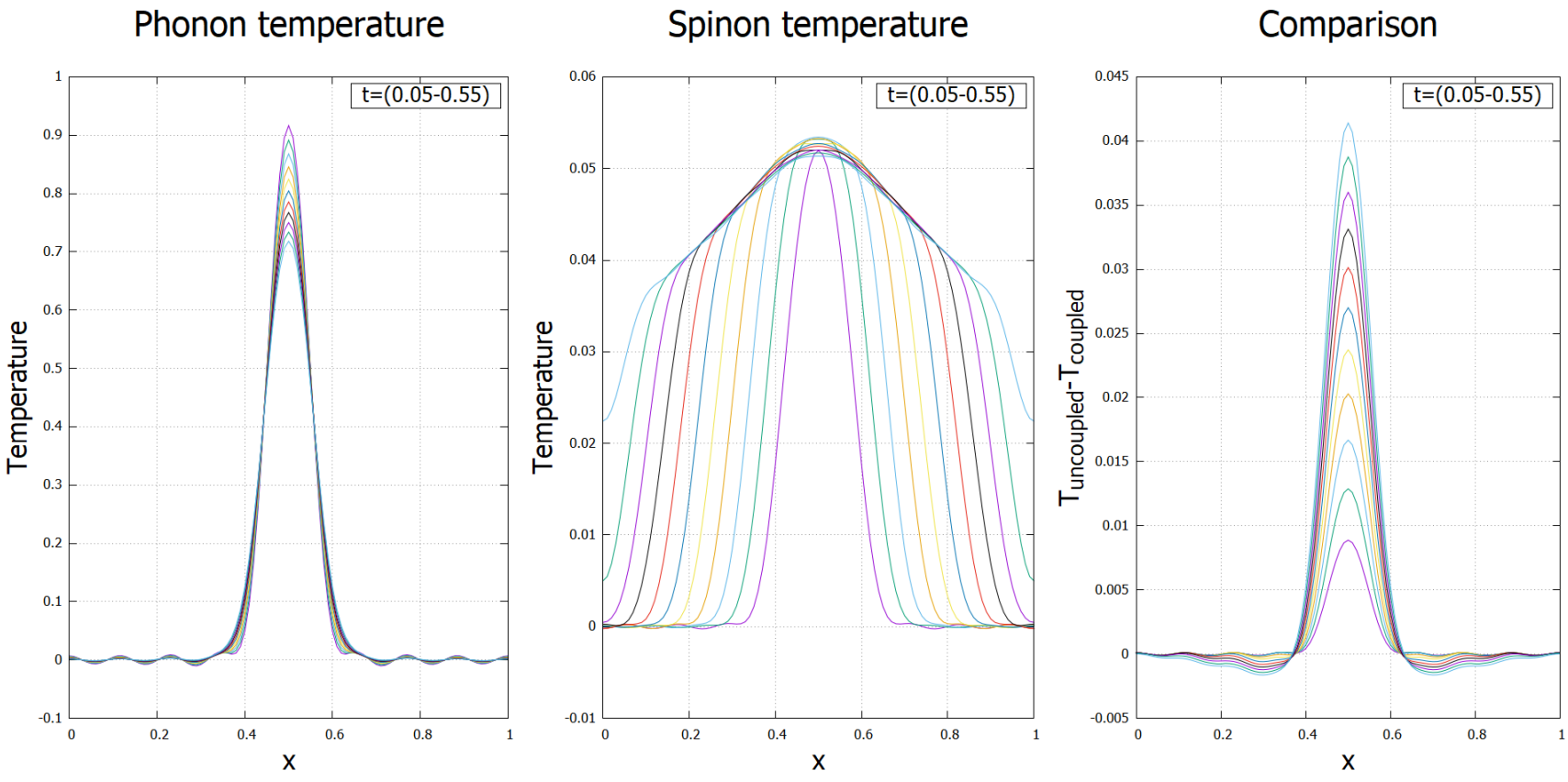

Phonon temperature
Spinon temperature
Comparison
t=(0.05-0.55)
t=(0.05-0.55)
t=(0.05-0.55)
Temperature
Temperature
T_uncoupled-T_coupled
x
x
x

Figure: $v = 0.8$

Varying D, $v = 0.5$, $c_p = 0.9$, $c_m = 0.1$

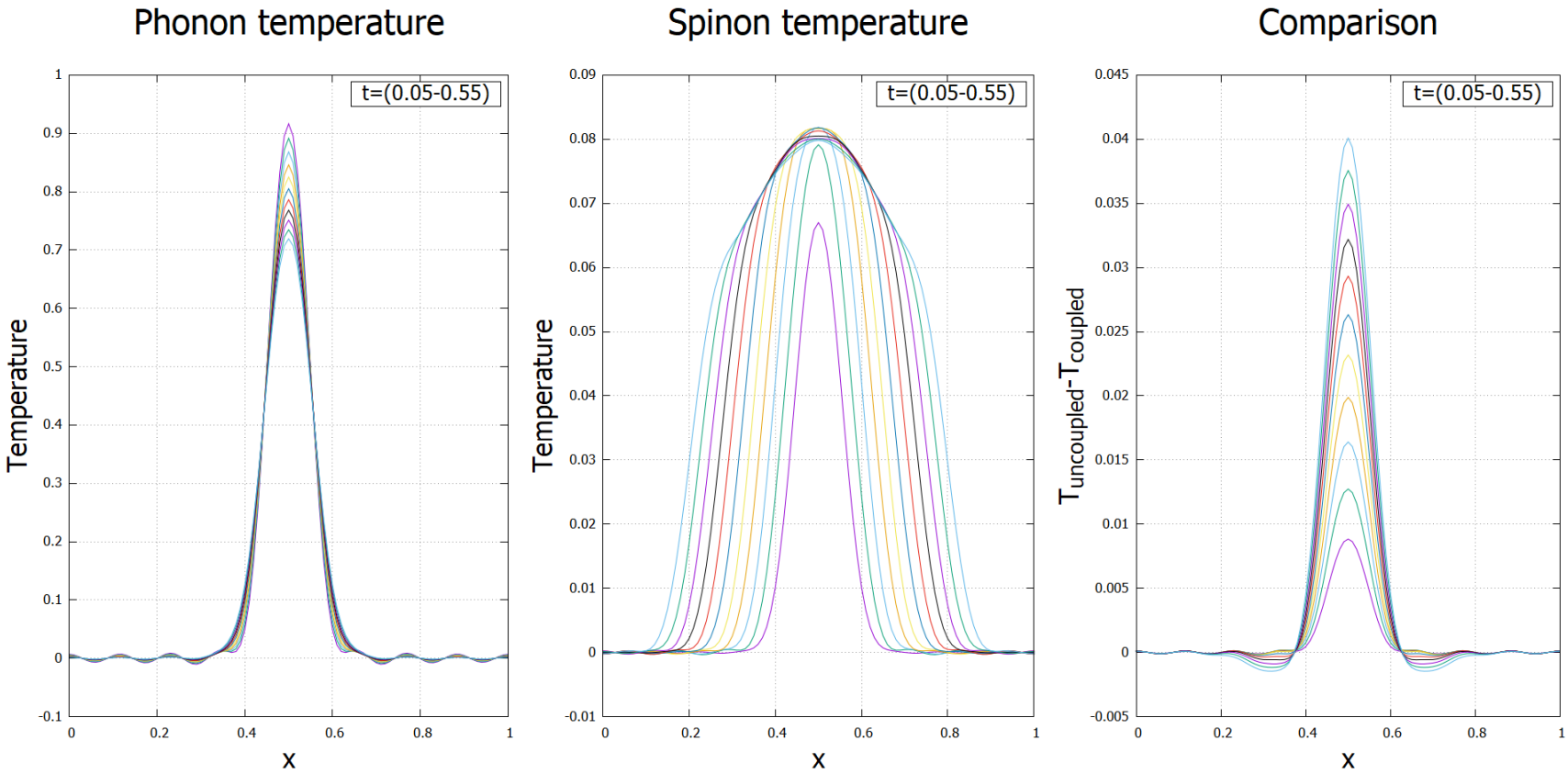


Figure: $D = 0.001$

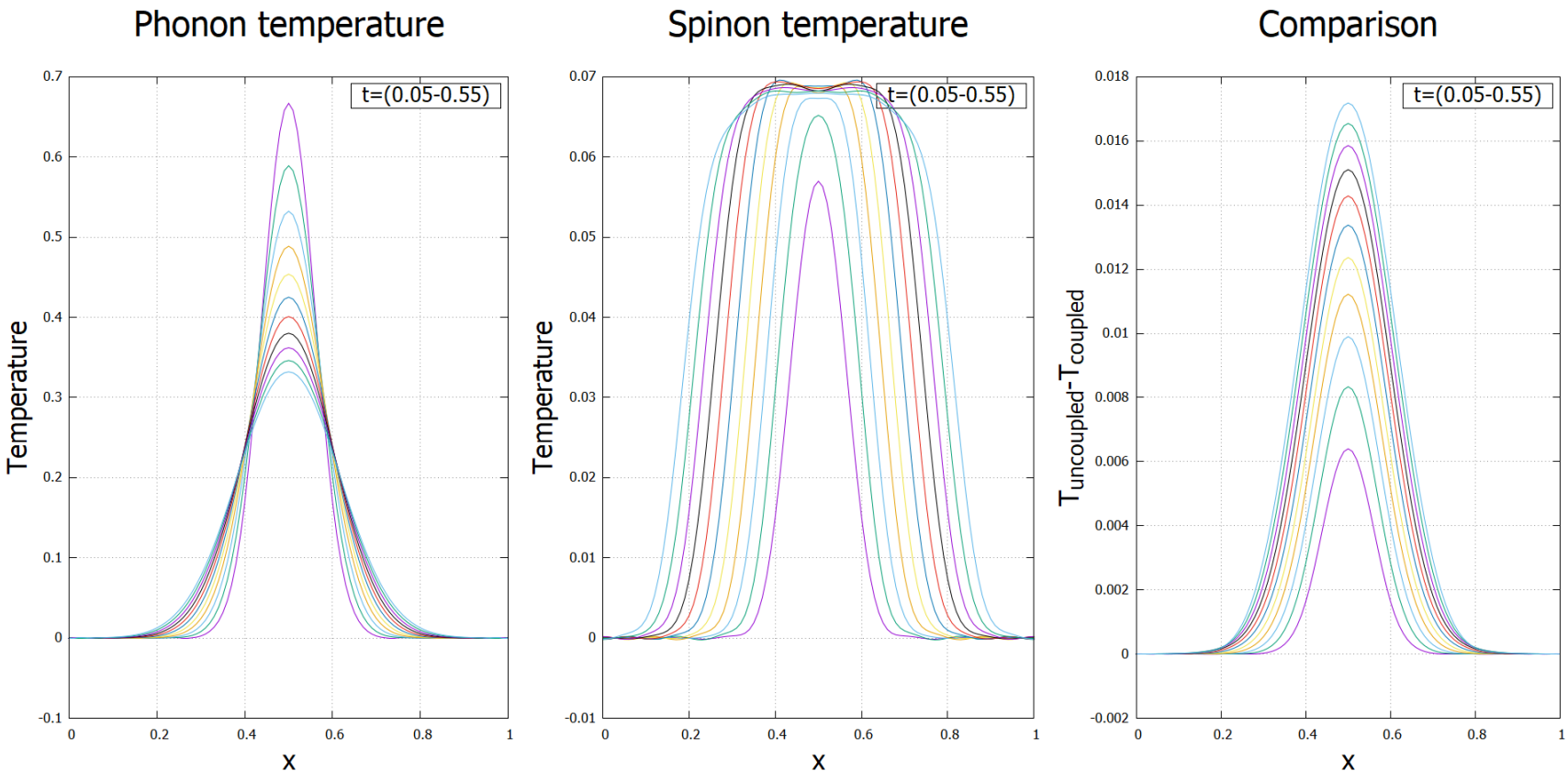

Phonon temperature
Spinon temperature
Comparison
t=(0.05-0.55)
Temperature
T_uncoupled-T_coupled
x

Figure: $D = 0.01$

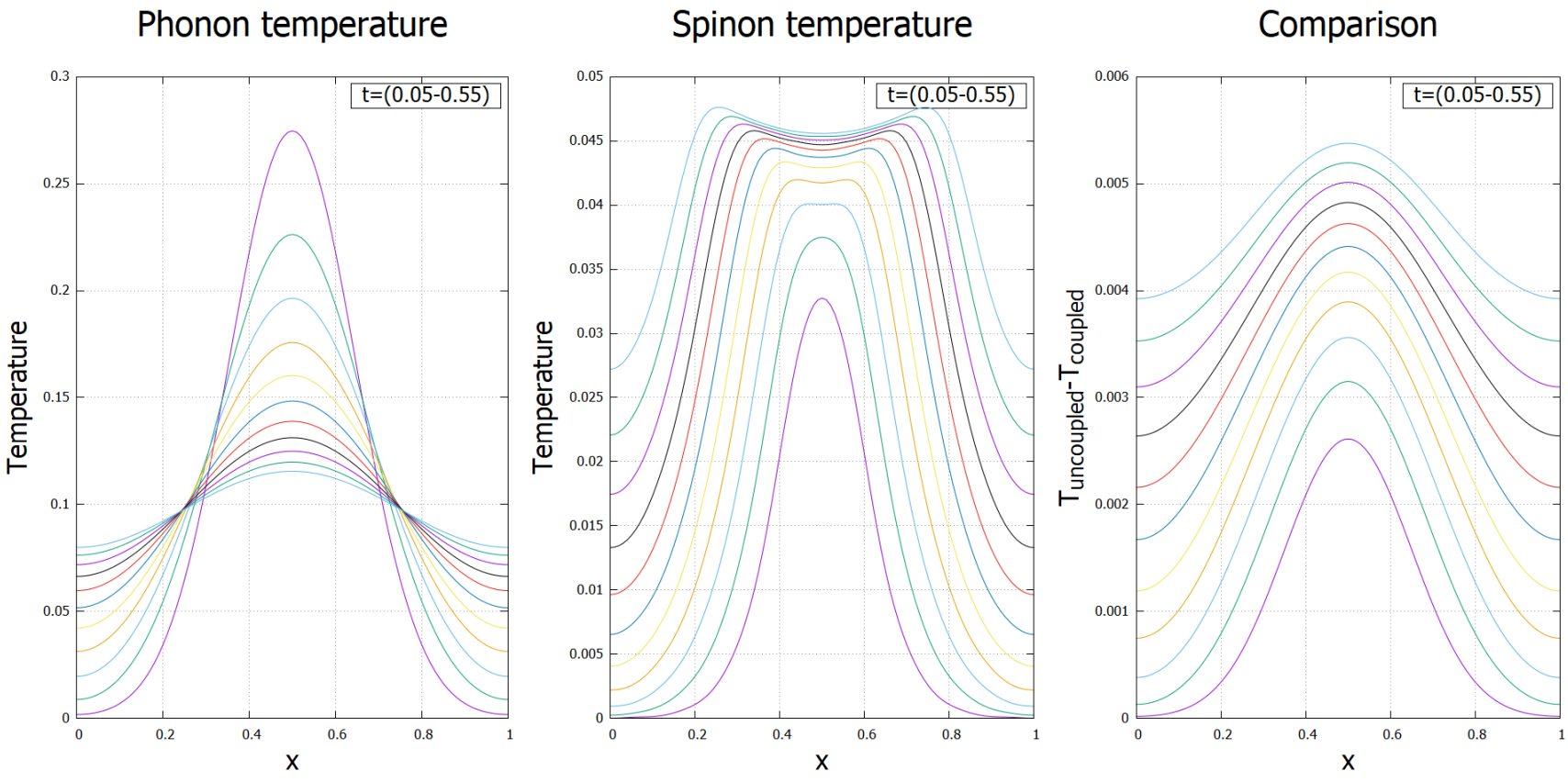

Phonon temperature
Spinon temperature
Comparison
t=(0.05-0.55)
t=(0.05-0.55)
t=(0.05-0.55)
Temperature
Temperature
T_uncoupled-T_coupled
x
x
x

Figure: $D = 0.1$

Varying c_p/c_m with $c_p + c_m = 1$, $D = 0.05$, $v = 0.8$

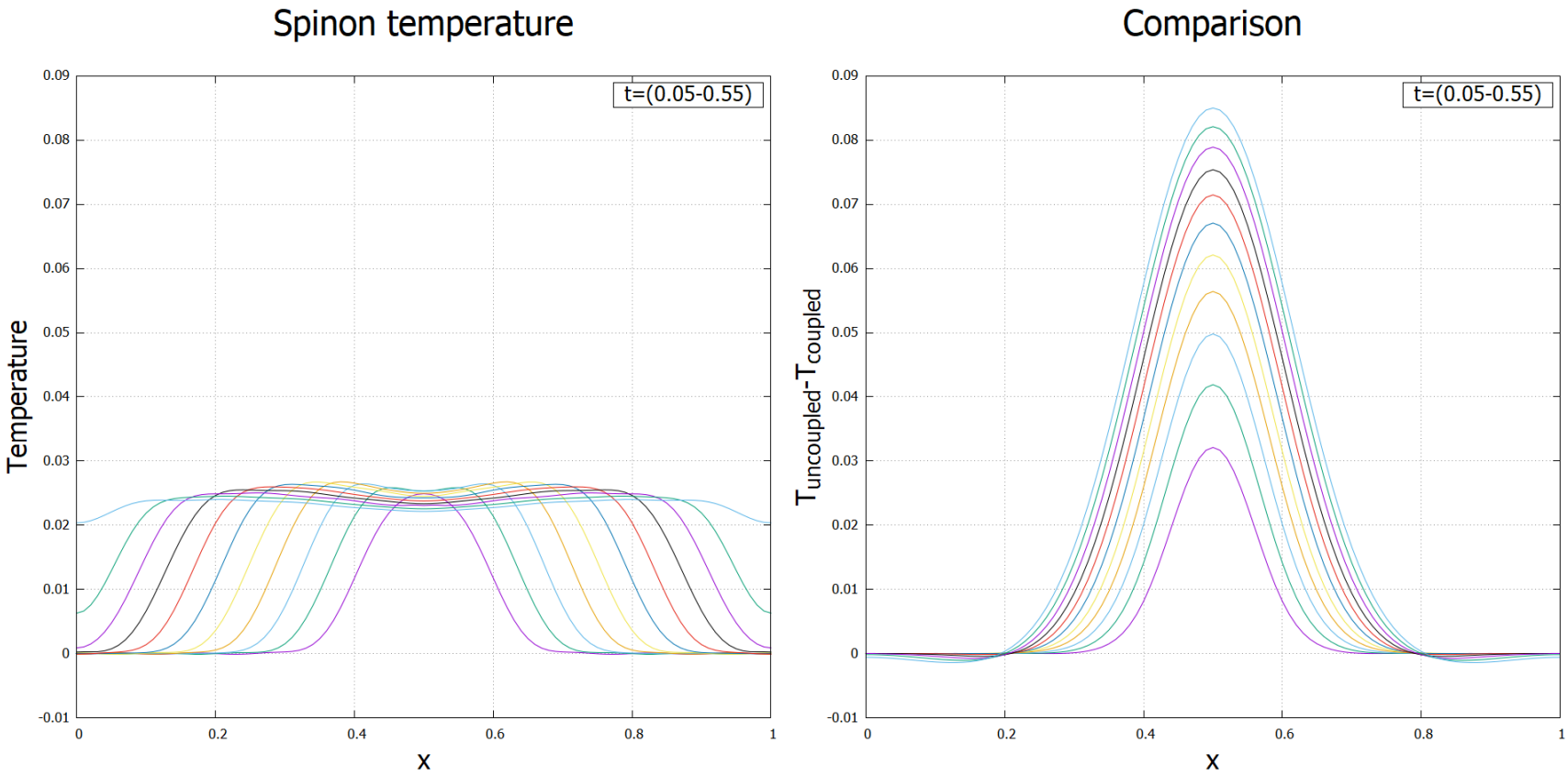


Figure: $c_p = 0.5, c_m = 0.5$



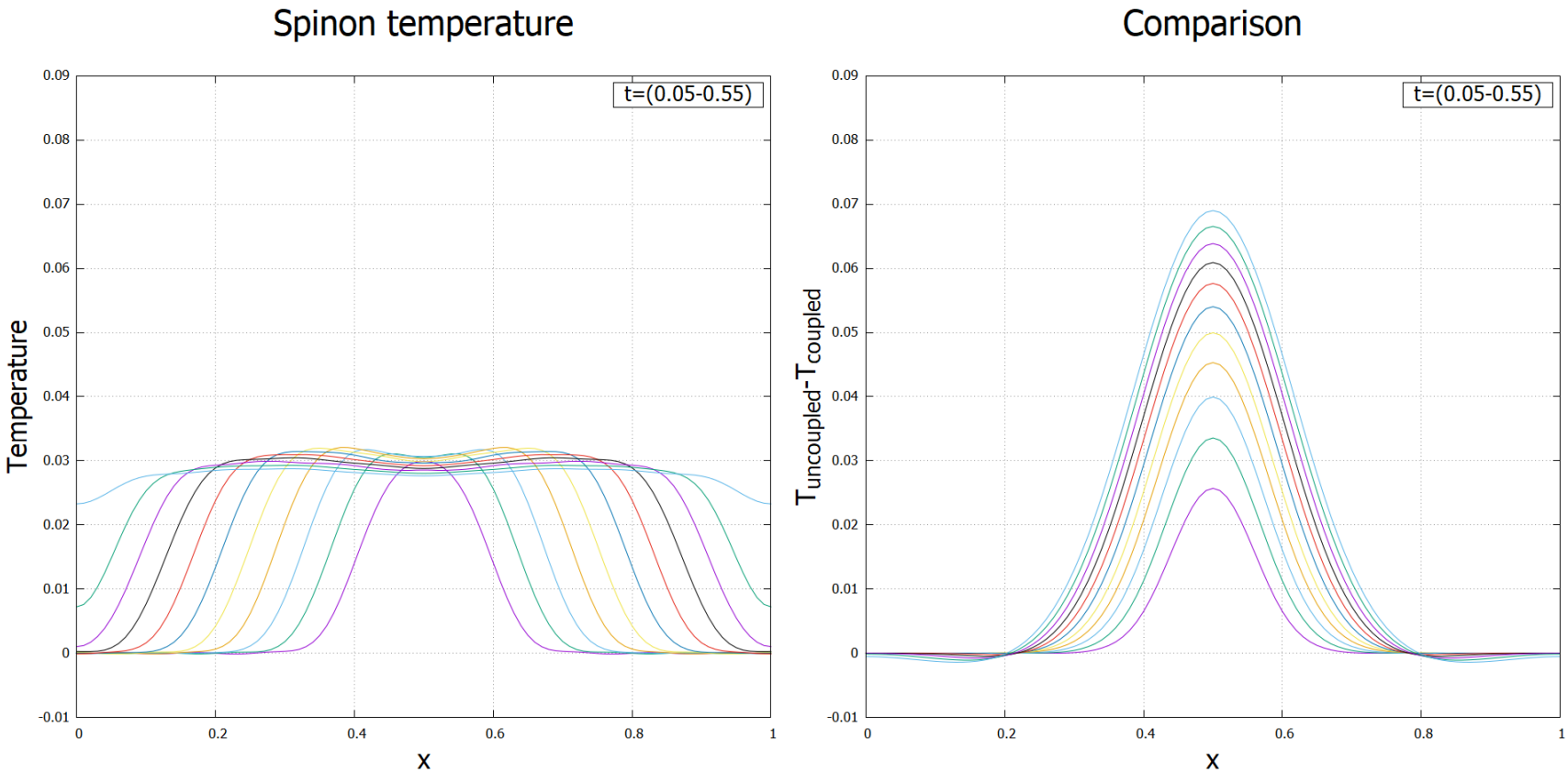


Figure: $c_p = 0.6, c_m = 0.4$

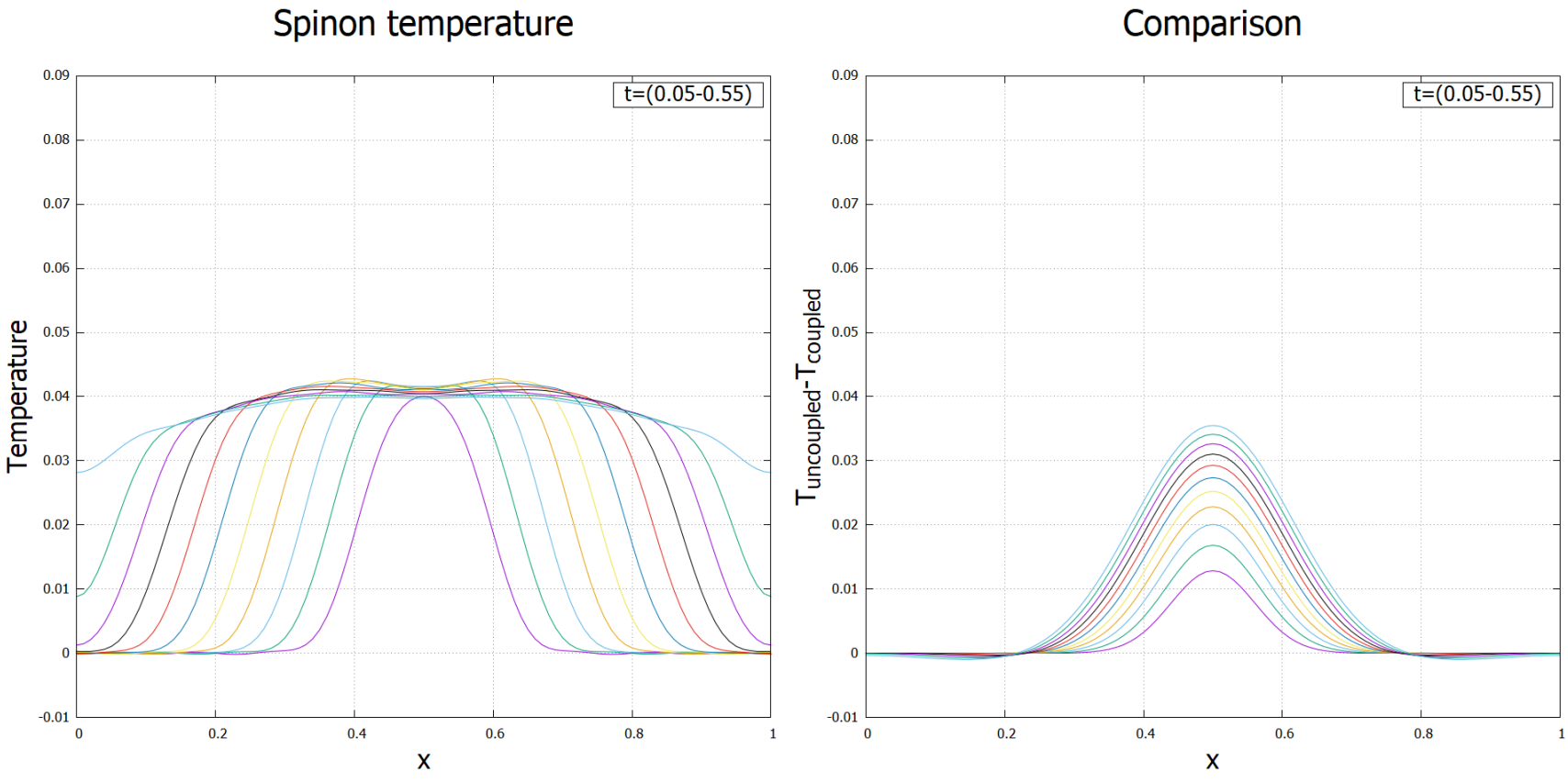


Figure: $c_p = 0.8, c_m = 0.2$



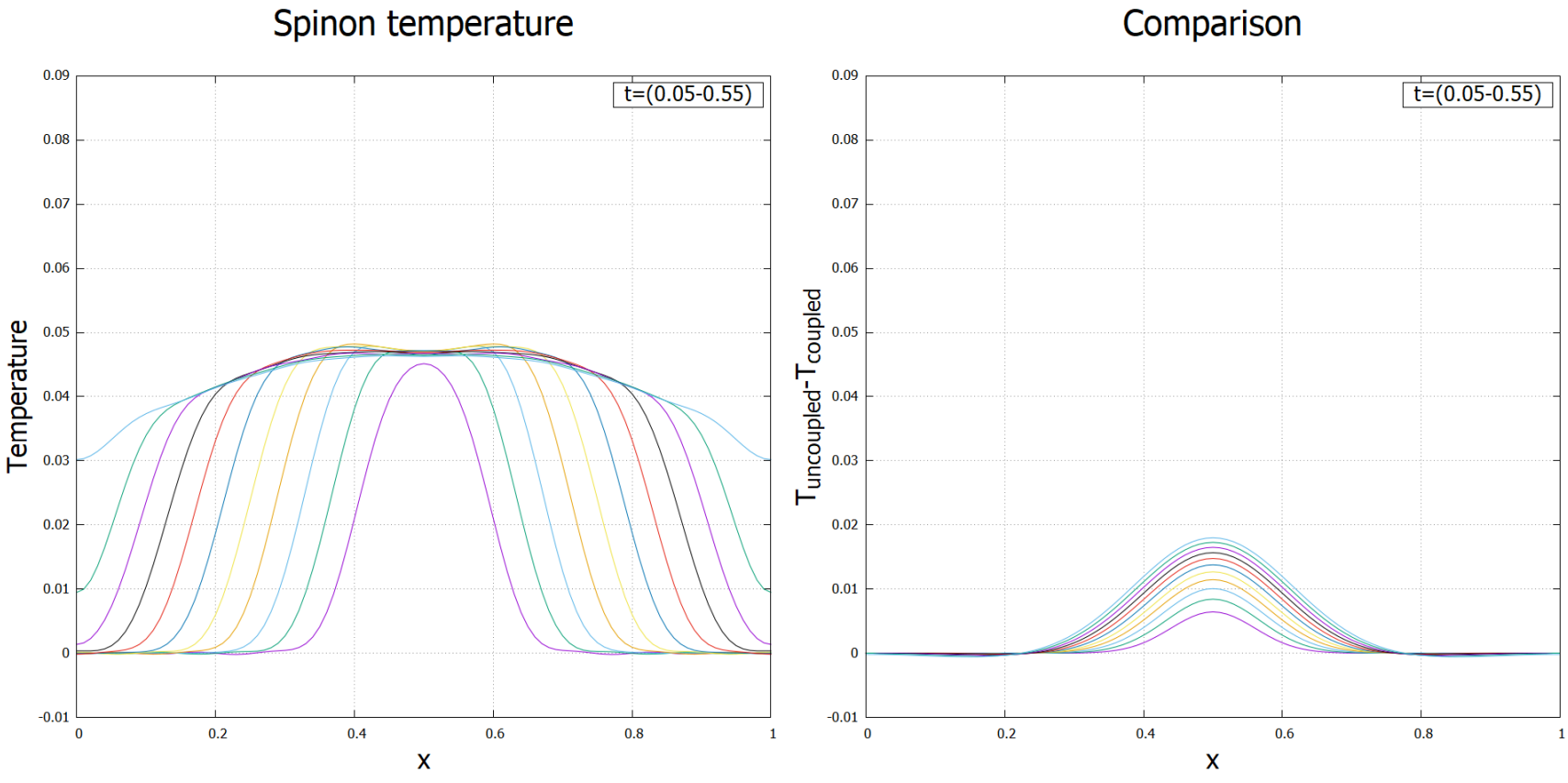


Figure: $c_p = 0.9, c_m = 0.1$



Experimental guidelines

Ballistic thermal transport is best seen when :

- D is small
- v is large
- $cp/cm \rightarrow 1$



Thank you.

